

Graph Theory $\rightarrow$ Foundations

A structured introduction for engineers (11 parts)

Discrete Maths for Engineers

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Part 1 Vertex & Edge

Definition

A **vertex** is an element of the set V . Vertices represent the objects of the modeled problem (cities, users, states, etc.).

Definition

An **edge** is a connection between vertices. In directed settings it is typically an ordered pair (u, v) . In undirected settings it is an unordered pair $\{u, v\}$.

Note

You can think of V as “things” and E (or A) as “relationships” between things.

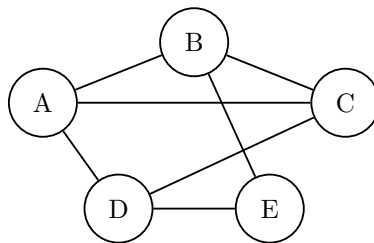


Figure 1: Example of an undirected graph with five vertices.

Part 2 Graph

Definition

A graph is a pair $G = (V, E)$, where V is a finite non-empty set of vertices and E is a set of edges connecting vertices.

Note

Different texts use different symbols for edges: E is common for undirected graphs, while A is also used (especially when emphasizing arcs in directed graphs).

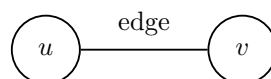


Figure 2: A minimal graph model: vertices and an edge.

Part 3 Graph (Example / Notation)

Example

Let $V = \{A, B, C, D, E\}$. An example edge set is:

$$E = \{\{A, B\}, \{B, C\}, \{A, C\}, \{A, D\}, \{C, D\}, \{B, E\}, \{D, E\}\}.$$

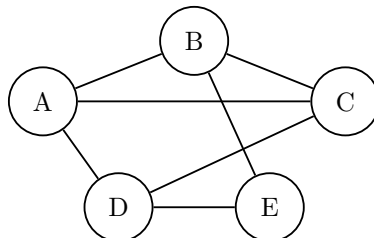


Figure 3: A concrete example of $G = (V, E)$ with $V = \{A, B, C, D, E\}$.

Part 4 Directed Graph

Definition

A **directed graph** (digraph) is $G = (V, A)$ with $A \subseteq V \times V$. Each arc (u, v) has orientation from u to v .

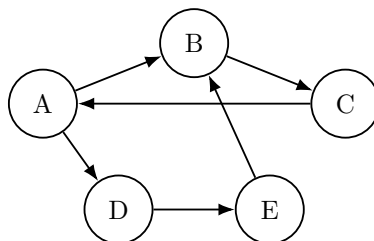


Figure 4: A directed graph (digraph): edges have orientation.

Part 5 Walk

Definition

A **walk** is a sequence $(s_0, s_1, \dots, s_k)$ such that $(s_i, s_{i+1}) \in A$ for all $0 \leq i < k$. The walk has length k .

Note

Walks may repeat vertices and edges.

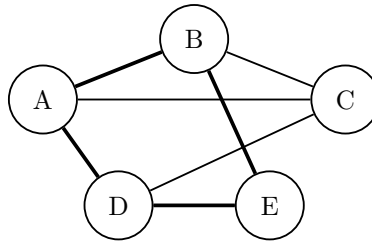


Figure 5: A walk highlighted: $A \rightarrow B \rightarrow E \rightarrow D \rightarrow A$.

Part 6 Closed Walk

Definition

A **closed walk** is a walk that starts and ends at the same vertex, i.e., $s_0 = s_k$.

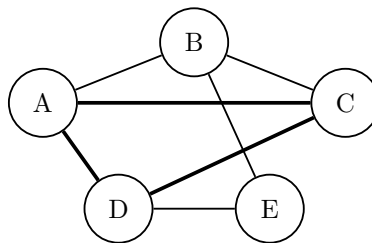


Figure 6: A closed walk highlighted: it starts and ends at A .

Part 7 Path

Definition

A **path** is a walk in which no vertex appears more than once.

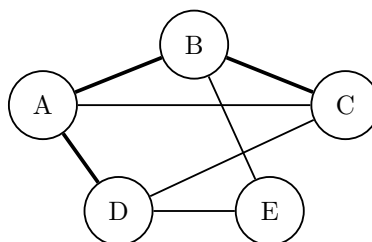


Figure 7: A path highlighted: $D \rightarrow A \rightarrow B \rightarrow C$.

Part 8 Circuit & Simple Circuit

Definition

A **circuit** is a closed walk that does not repeat edges.

Definition

A **simple circuit** is a circuit with no repeated vertices, except the first vertex equals the last.

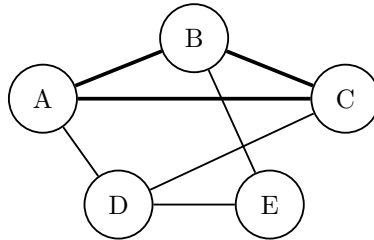


Figure 8: A simple circuit highlighted: $A \rightarrow B \rightarrow C \rightarrow A$.

Part 9 Cycle

Definition

A **cycle** is a closed walk in which no vertex is repeated, except the initial vertex, which coincides with the final vertex.

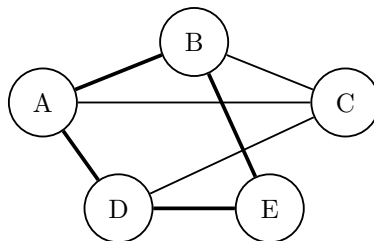


Figure 9: A cycle highlighted (no repeated vertices except start/end).

Part 10 Connected Graph

Definition

An undirected graph $G = (V, E)$ is **connected** if for every $u, v \in V$ there exists a path between u and v .

Definition

A directed graph is **strongly connected** if for every $u, v \in V$ there exists a directed path from u to v and from v to u .

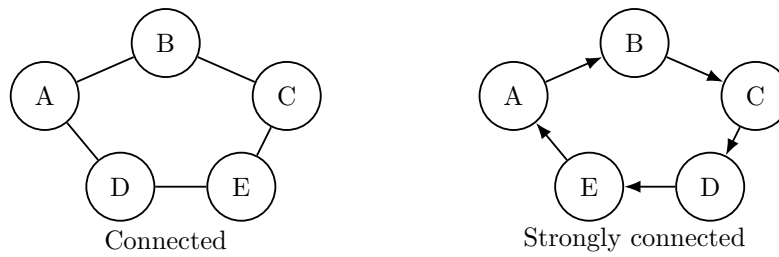


Figure 10: Connectedness in undirected graphs vs. strong connectivity in directed graphs.

Part 11 Connected Component

Definition

A **connected component** of an undirected graph is a maximal connected subgraph.

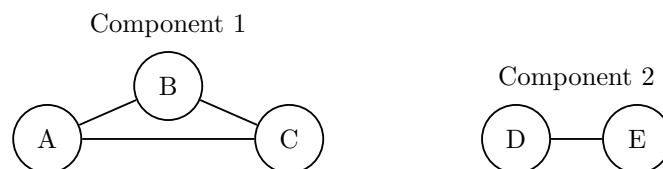


Figure 11: A graph with two connected components (disconnected overall).

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